Assignment 2: Secret Codes (Arrays & Functions)

Assignment for COSC 1100 – Intro to Programming

Prior to attempting this problem, you should have done the following:
1. View the Content, including assigned readings for weeks 5 through 9.
2. Viewed the Functions in Python video from week 9 and completed the associated Pre-Class Activity.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Analyze the problem, design a documented plan (i.e. flowchart or pseudo-code), code and test a solution **following the step-by-step approach** presented in this course.
3. Submit your solution and plan to the appropriate assignment folder on DC Connect **by the due date provided**.
4. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

Create a simple Python application following the requirements below to allow a user to encode a piece of text using a cipher, or to decode the text using the same cipher. A "cipher" is a method of making a coded message that can be "deciphered" – usually the input to a cipher is the message to encode and some kind of "key" value that makes it make sense.

In your program, after the user picks whether they are encoding or decoding the text, the user will enter text and it will either be encoded or decoded. Since your program allows both encoding and decoding, you will be able to (*partly*) test if the cypher works using just your own program.

Detailed Requirements:

- **Initial prompt**: When the program starts the user is prompted to enter "1" to encode, "2" to decode, or "3" to exit.
- If the user inputs "1":
    - Accept one text input from the user; validation of this input may not be necessary
    - Pass the user's text input to an "encode" function
    - Print the encoded text string that was returned by the "encode" function
    - Do the **initial prompt** again
- If the user inputs "2":
    - Accept one text input from the user; validation of this input may not be necessary
    - Pass the user's text input to an "decode" function
    - Print the decoded text string that was returned by the "decode" function
    - Do the **initial prompt** again
- If the user inputs "3":
    - End the program
- If the user inputs anything other than "1", "2" or "3", re-prompt them with the **initial prompt** again.

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Requirements for the Cipher Functions:

In this case, we will be using a Substitution cipher, which is a very simple type of cipher that operates by replacing characters from one "alphabet" with the letters of another "alphabet". In our case, we will convert from English letters into the Al Bhed language from Final Fantasy X. Your encode and decode functions will each have two parameters: the message, and a key value, where the key value represents how many times the substitution cipher will run. The key will have a default value of 1.

English	Al Bhed
A	Y
B	P
C	L
D	T
E	A
F	V
G	K
H	R
I	E

English	Al Bhed
J	Z
K	G
L	M
M	S
N	H
O	U
P	B
Q	X
R	N

English	Al Bhed
S	C
T	D
U	I
V	J
W	F
X	Q
Y	O
Z	W

There are many viable ways to implement a Substitution cipher in Python, but probably the easiest way for our needs would be to declare two strings that include each of the two entire alphabets. The English alphabet could be "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz" and the Al Bhed alphabet would be similar but re-arranged based on the substitution table above. Then, given a coded message, for each letter, match that letter's position in the first alphabet with the same position in the second alphabet.

Here is some partial pseudocode for the encode function that you can make use of:

```
Start with an empty string called encrypted_string
For each letter_index, counting up from 0 to the length of the input
    Assume that there is no match found (match_found = False)

    For each alphabet_indx in the alphabet
        If the letter at letter_index matches the letter at alphabet_index
        Add that letter to the encrypted_string
        Set match_found to True
    If after going through the whole alphabet, match_found is False
        Add the original letter at letter_index to the encrypted string
```

Notably omitted from this code snippet is the key value. The entire substitution algorithm should run a number of times equal to the key (which has a default value of 1). Each additional time it is encoded will "scramble" it differently, and decoding it the same number of times will *un*scramble it.

Style Guide:

To be eligible for full marks on this or any assignment in this course your application must conform to the requirements as outlined above as well as our prescribed style guide, in this case making sure to observe the PEP8 naming conventions for Python as well as appropriate and complete program documentation.

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Development Hints:

- As always, start with your planning steps. The pseudocode shown above may be part of the process and plan (steps 3 and 4 of 10).
- Only *after* planning, consider how you can "translate" the given pseudocode into Python code.
- Ciphers are a very common programming exercise and thus there are numerous reference materials and possibly even full solutions available on the web. However, be cautious with these!
    - Make sure any "borrowed" code *works* and that your program actually meets our requirements.
    - Make sure all code in your program conforms to our PEP8 style guide.
    - Reference any code directly taken from an online resource or another person, or this will be considered an incident of plagiarism. You can just do this in a comment before the part you have taken from another source.

Sample Output:

```
1) Encode text
2) Decode text
3) Exit program
Please select an option (1-3): 1
Please enter the text to encode: Hello, it's nice to meet you.
Encoded text: Rammu, ed'c hela du saad oui.

1) Encode text
2) Decode text
3) Exit program
Please select an option (1-3): 3.1

Please select a valid option between 1 and 3.

1) Encode text
2) Decode text
3) Exit program
Please select an option (1-3): 2
Please enter the text to decode: Rammu, ed'c hela du saad oui.
Decoded text: Hello, it's nice to meet you.
```

Academic Integrity

By submitting this assignment to DC Connect for evaluation, you agree to the following terms:

- ☐ This work is your original work, with the majority its contents originally created by yourself.
- ☐ Any aspects of this work that were created by or based heavily on reference to another person's work, a website, a video, or a program (including generative AI systems) are cited with suitable identification to observe the original source.
- ☐ You did not transmit this file to anyone else's computer, nor receive any part of this work from anyone else's computer.

Failure to observe these points will leave you subject to Durham College's Academic Integrity Policy, with a minimum penalty of a grade of 0 on the assignment and an academic integrity alert submitted to the faculty office.